

Name _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Class Reg Number

17	
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Calculator Model / No.

Chemistry

8873/02

Paper 2

13 September 2018
2 hours

Additional Material: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces at the top of this page.

This booklet contains Section **A** and Section **B**.

Section A (60 marks): Page 2 to 15

Answer **all** questions in the spaces provided. You are advised to spend about **1 h 30 min** on Section **A**.

Section B (20 marks): Page 16 to 24

Answer **one** question in Section **B** in the spaces provided. You are advised to spend about **30 min** on Section **B**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Examiner's Use		
Paper 1	MCQ	/ 30
	/ 33 %	
Paper 2 Section A	Q1	/ 15
	Q2	/ 17
	Q3	/ 16
	Q4	/ 12
Paper 2 Section B	/ 20	
Paper 2 Total	/ 80	
	/ 67 %	
Grand Total	/ 100 %	
Grade		

This document consists of **24** printed pages. (inclusive of this cover page)

Section A

Answer **all** questions in this section, in the spaces provided.

- 1 Chemists have long studied the trend of elements, and the Periodic Table was celebrated as an effective way to arrange the elements while reflecting some of the most well-known trends in their physical and chemical properties.
 - (a) Using structure and bonding, describe and explain the variation in melting points for elements aluminium to phosphorous.

[4]

[illegible]

- (b) (i)** The ionisation energies of Period 3 elements from sodium to argon generally increases, with the exception of two decreases: from Mg to Al, and from P to S.

Explain these two anomalies in the trend.

[2]

From Mg to Al

From P to S

- (ii) An element from Period 3 was found to have the following successive ionisation energy values. Deduce the identity of this element.

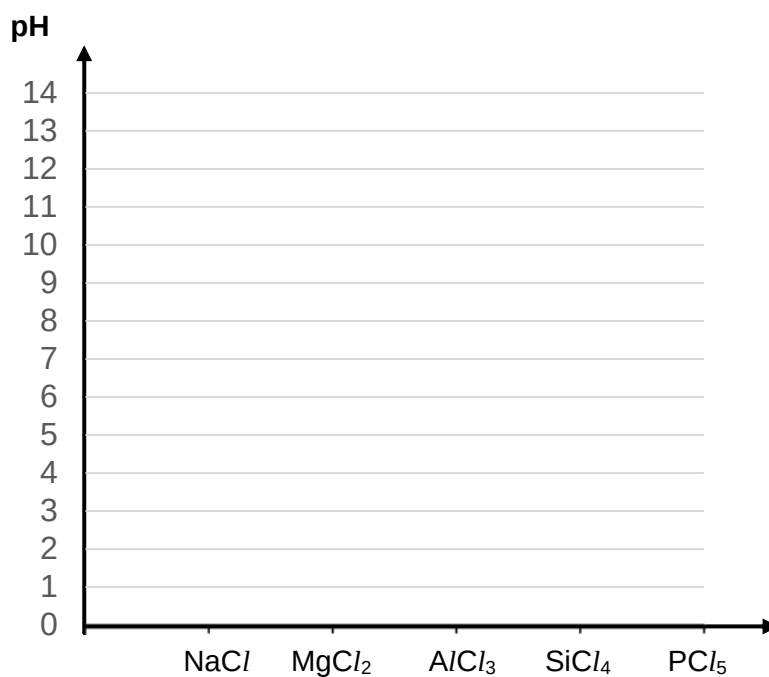
Ionisation Energy	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Value / kJ mol^{-1}	790	1600	3200	4400	16000	20000	24000	29000

[1]

- (c) Chlorides of Period 3 elements display a wide range of properties.

- (i) The pH of the resultant solution when Period 3 chlorides dissolves in or react with water shows a periodic trend. Sketch the pH trend from NaCl to PCl_5 .

[1]



- (ii) Write an equation for the reaction of PCl_5 with water.

[1]

2(a) Nearly all petrol and diesel vehicles have some kind of catalytic converter fitted to the exhaust to reduce the amount of pollutants emitted into the environment. The catalytic converter usually contains palladium nanoparticles which act as catalyst.

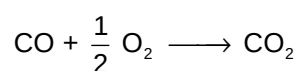
(i) Define the term nanoparticles.

[1]

(ii) Explain why palladium used is in the form of nanoparticles instead of its bulk form.

[2]

(iii) The palladium catalyst helps to remove carbon monoxide from the exhaust. For example,

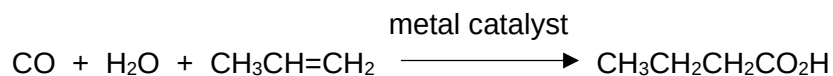


State the **type** of catalyst and outline the mode of action of catalyst in this reaction.

[2]

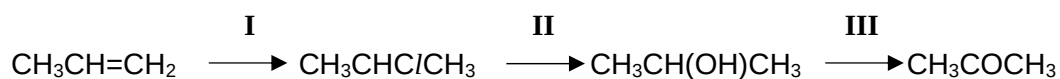
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- (b) Carbon monoxide, water and propene can be used to produce butanoic acid according to the following equation:



Draw the structure of the product formed when 2-methylpropene is used in the above reaction. [1]

- (c) Propene can be converted to propanone by the following route.



- (i) Suggest the types of reaction for steps I and III. [2]

	step I	step III
type of reaction		

- (ii) State the reagent and conditions for step II. [1]

.....

- (iii) State and explain how the rate of reaction for step II changes when 2-chloropropane is replaced by 2-iodopropane [2]

.....

.....

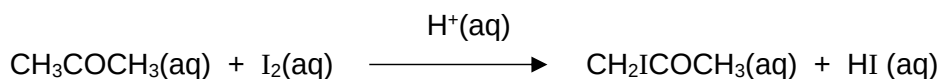
.....

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- (d) The kinetics of the acid-catalysed reaction of propanone with iodine



can be investigated experimentally by varying the concentrations of the three substances involved and determining the time for the colour of the iodine to disappear.

In this method the rate of the reaction is measured in terms of the rate at which the iodine changes, i.e.

$$\text{rate} \propto \frac{\text{volume of aqueous iodine used}}{\text{time for the colour of iodine to disappear}}$$

The following results were obtained in such an investigation.

Expt	Volume of CH_3COCH_3 / cm^3	Volume of I_2 / cm^3	Volume of H^+ / cm^3	Volume of H_2O / cm^3	Relative time for the colour of iodine to disappear
1	8	4	8	0	1
2	8	4	4	4	2
3	4	4	8	4	2
4	8	2	8	2	0.5

- (i) Deduce the order of reaction with respect to propanone, iodine and hydrogen ions.

[3]

propanone

iodine

hydrogen ions

- (ii) Hence write the rate equation for this reaction.

[1]

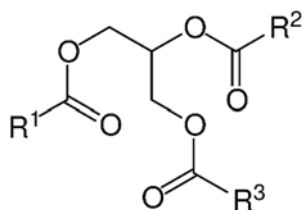
- (iii) Another experiment, with 0.80 mol dm^{-3} propanone, $0.0010 \text{ mol dm}^{-3}$ iodine and 0.20 mol dm^{-3} hydrochloric acid, is carried out at the same temperature.

Calculate the value of the rate constant if the rate is $4.2 \times 10^{-6} \text{ mol dm}^{-3} \text{ s}^{-1}$. Give the units of the rate constant.

[2]

[Total: 17]

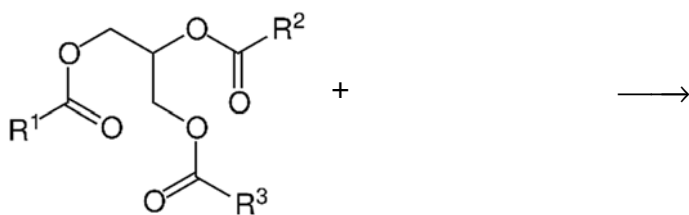
- 3(a) Limited scientific evidence suggests that eating two tablespoons of olive oil daily may reduce the risk of coronary heart disease. Olive oil comprises mainly of the mixed triglyceride esters of oleic acid, palmitic acid and other fatty acids. The general chemical structure of olive oil is shown below where R^1 , R^2 and R^3 are alkyl groups or alkenyl groups.



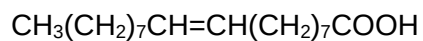
Olive oil undergoes rancidification when exposed to moisture, resulting in an unpleasant taste and odour. The products formed in the reaction are propane-1,2,3-triol and fatty acids.

Complete the equation to illustrate the rancidification reaction of olive oil.

[1]



(b) Oleic acid has the structure shown.



- (i) Describe the bonding in C=C in terms of orbital overlap. Draw a diagram to illustrate your answer.

[2]

- (ii) Oleic acid exists as a pair of isomers.

Draw and label the structural formulae of the two isomers.

[2]

- (iii) Describe a chemical test that would distinguish oleic acid from ethanoic acid.

[2]

- (c) **Fig. 3.1** shows a bomb calorimeter, inside a controlled temperature water jacket, which is used to find an accurate value of standard enthalpy change of combustion of oleic acid.

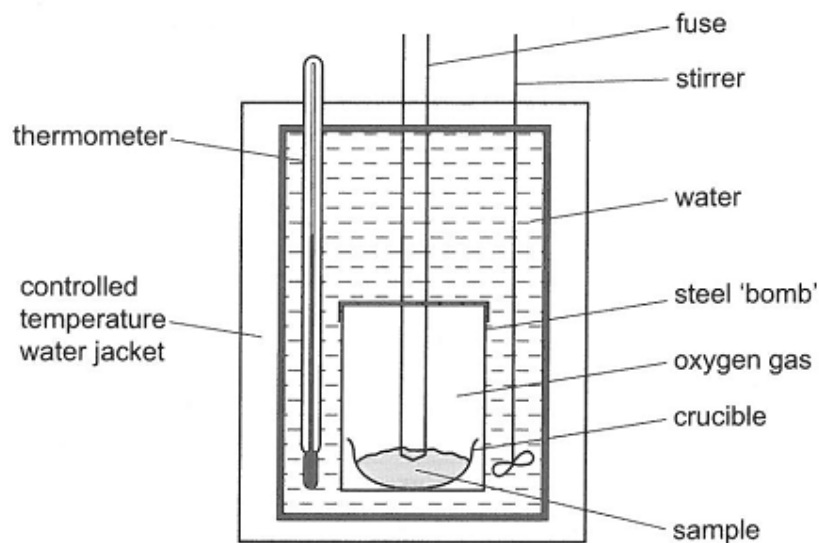


Fig. 3.1

- (i) What is meant by the *standard enthalpy change of combustion* of oleic acid?

[1]

- (ii) Explain why the steel 'bomb' is flushed with a high pressure of oxygen at the start of the experiment.

[1]

- (iii) There is no heat lost from the calorimeter because of the controlled temperature water jacket.

Suggest how the controlled temperature water jacket achieves this.

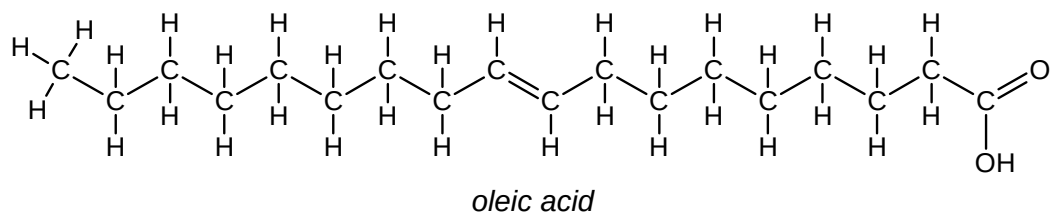
[1]

- (iv) A 2.98 g sample of oleic acid ($C_{18}H_{34}O_2$) is burned in a bomb calorimeter with a heat capacity of 1.98 kJ K^{-1} . The temperature of the calorimeter increases by 59.3°C .

Calculate the standard enthalpy change of combustion of oleic acid.

[2]

- (v) Use the bond energies given in the *Data Booklet* to calculate another value for the standard enthalpy change of combustion of oleic acid.



You may find the following information useful in your calculation.

An oleic acid ($C_{18}H_{34}O_2$) molecule contains 33 C–H bonds, 16 C–C bonds and other bonds.

[3]

- (vi) Suggest a reason for the discrepancy between the value in (c)(iv) and that in (c)(v).

[1]

[Total: 16]

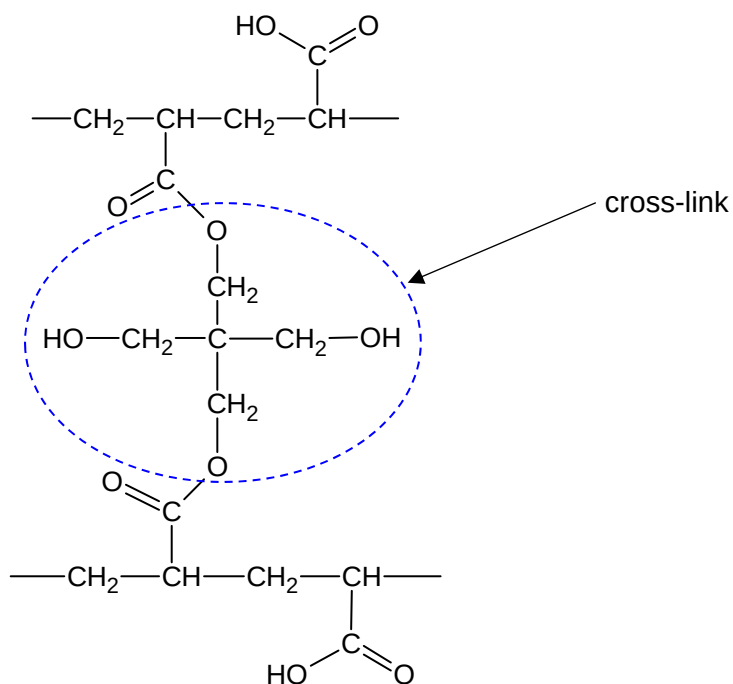
4(a) There are two types of polymerisation, addition and condensation

Account for the differences between these two types of polymerisation.

[2]

(b) Super-absorbent polymers have the ability to absorb 200 – 300 times their own mass of water. They are classified as hydrogels and are widely used in personal disposable hygiene products such as diapers. However, these synthetically made hydrogels are usually non-biodegradable.

The diagram below shows part of the structure of a hydrogel.



The hydrogel is formed from chains of one polymer which are then cross-linked using another molecule.

(i) State the type of polymerisation used to form these polymer chains.

[1]

- (ii) Draw the **displayed** structural formula of the monomer used.

[1]

- (iii) Draw the structure of the molecule used to cross-link the polymer chains.

[1]

- (iv) Once a hydrogel has absorbed water, it can be dried and re-used many times. Explain why this is possible, referring to the structure.

[1]

- (v) The amount of cross-linking has little effect on the ability of the gel to absorb water. Suggest why this is the case.

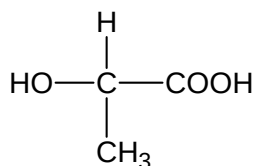
[1]

- (vi) Suggest one property of the hydrogel that will change if more cross-linking takes place. Explain how the increased cross-linking brings about this change.

[2]

- (c) Unlike hydrogel, the polymer, 'polylactide' or PLA is biodegradable. The monomer required to produce PLA is *lactic acid*, 2-hydroxypropanoic acid.

The structure of *lactic acid* is as shown.



- (i) Draw the structure of the polymer PLA, showing **two** repeat units.

[1]

One of the reasons PLA has attracted so much attention is that it is biodegradable. This does, however restrict some potential uses. The polymer has a melting point of around 175 °C , but soften between 60 – 80°C. However, its properties enable it to have a range of uses in food packaging and in fibres.

- (ii) Explain why PLA would not be a suitable packaging materials for food pickled in vinegar.

[1]

- (iii) Suggest why PLA containers are not usually used to hold hot drinks.

[1]

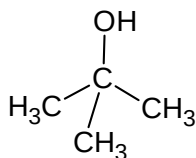
[Total: 12]

END OF SECTION A

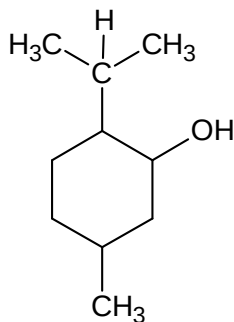
Section B

Answer **one** question from this section, in the spaces provided.

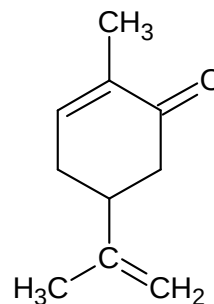
5(a) Three natural-occurring compounds used commercially are shown below:



2 methylpropan-2-ol



Menthol



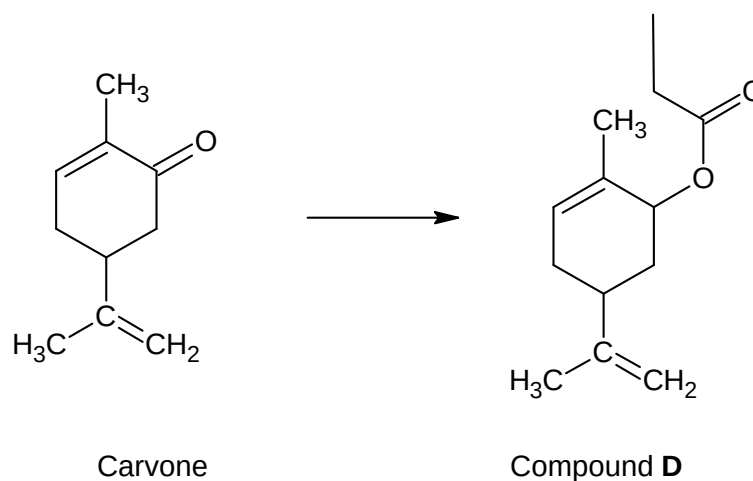
Carvone

(i) Describe a set of two chemical tests that can be used to distinguish the above three compounds. Write an equation for each reaction that takes place.

[5]

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- (ii) Carvone can be converted to organic compound **D** in two steps.

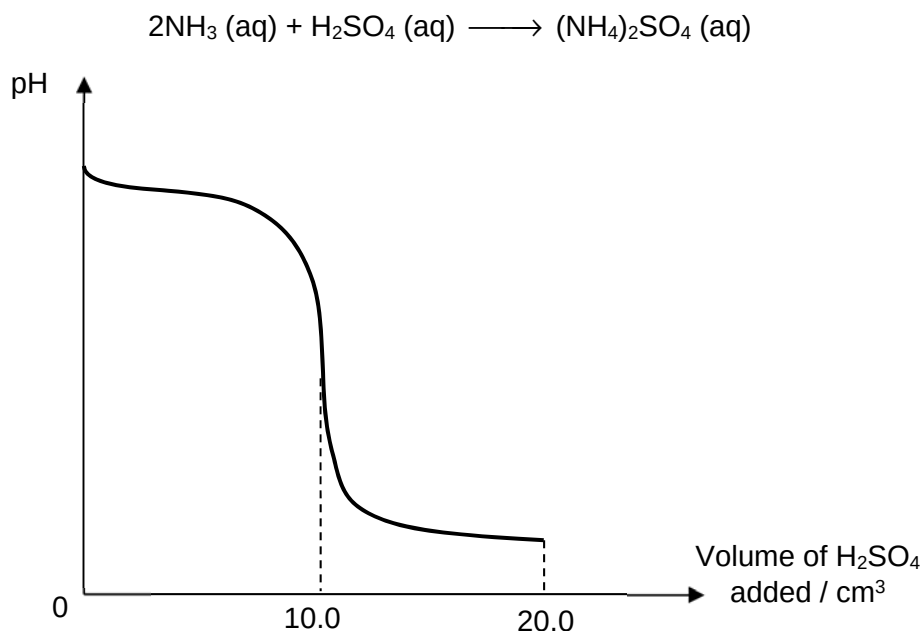


State the reagents and conditions necessary for each step. Give the structure of the intermediate involved in the conversion.

[3]

- (b) Ammonia was used in smelling salts to treat fainting. Modern smelling salt solution may contain other products such as lavender oil to act in conjunction with aqueous ammonia.

When 40 cm³ of aqueous ammonia was titrated against 0.200 mol dm⁻³ sulfuric acid, H₂SO₄ using a pH meter, the following graph was obtained.



- (i) Explain what is meant by the terms *Bronsted–Lowry base* and *conjugate acid–base pair*. Illustrate your explanation using ammonia in the above reaction with sulfuric acid.

Explain why the *Arrhenius* definition of base does not apply for ammonia in the above reaction.

[3]

- (ii) Write an expression for the base dissociation constant, K_b , of ammonia.

[1]

- (iii) Using the titration curve provided, state the volume of sulfuric acid needed for the exact neutralisation of 40 cm³ of aqueous ammonia. Hence, calculate the initial concentration of aqueous ammonia.

[2]

- (iv) The resultant solution at the end of the titration can be considered as a sulfuric acid solution of concentration 0.0333 mol dm⁻³. Calculate the pH of this resultant solution to 2 decimal place.

[1]

- (v) Suggest an appropriate indicator for the titration, giving a reason for your choice.

[1]

- (vi) Explain what do you understand by the term *buffer solution*.

[1]

- (vii) The *Henderson–Hasselbalch* equation is useful for the estimation of the pH of a buffer solution. An alternative form of the equation is shown below which can be used to determine the pOH of a basic buffer solution.

$$\text{pOH} = \text{p}K_{\text{b}} + \log_{10} \frac{[\text{BH}^+]}{[\text{B}]}$$

Calculate the pH of a basic buffer solution, if the solution contains NH_4^+ and NH_3 in a 2 : 1 ratio, given that the base dissociation constant, K_{b} of ammonia, NH_3 is $1.78 \times 10^{-5} \text{ mol dm}^{-3}$.

[2]

- (viii) The enthalpy changes of neutralisation involving sulfuric acid with two bases are shown below.

sodium hydroxide, NaOH	– 57.3 kJ mol ^{–1}
ethylamine, CH ₃ CH ₂ NH ₂	– 51.5 kJ mol ^{–1}

Comment and explain for the above observation as fully as you can.

[1]

[Total: 20]

- 6(a)** Hydroxylamine, NH_2OH is commonly used in photography developing solution. Prolonged exposure to hydroxylamine by photography developers might cause irritation to their respiratory tract. To minimise the health hazard on these developers, the safety limit of the concentration of hydroxylamine is capped at below $0.00200 \text{ mol dm}^{-3}$.

A scientist performed the following experiment to determine if a sample of photography developing solution was within the safety limit.

100 cm^3 of NH_2OH solution was added to 40 cm^3 of $0.015 \text{ mol dm}^{-3}$ acidified $\text{I}_2 (\text{aq})$ solution. In this reaction, iodine is used in excess and NH_2OH is oxidised to form dinitrogen monoxide, N_2O .

- (i)** Using half-equations, write a balanced equation for the reaction between $\text{NH}_2\text{OH} (\text{aq})$ and $\text{I}_2 (\text{aq})$ in an acidic medium.

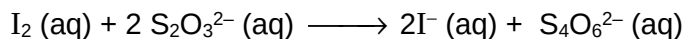
[2]

Reduction: _____

Oxidation: _____

Overall: _____

A 25.0 cm^3 aliquot of the resultant reaction mixture was then titrated with a standard solution of $0.00500 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3 (\text{aq})$.



Starch solution was added as indicator for this titration to enhance the contrast of the colour change at the end-point of the titration. 22.50 cm^3 of $\text{Na}_2\text{S}_2\text{O}_3 (\text{aq})$ was required.

- (ii)** State the expected colour change at the end-point of the titration.

[1]

- (iii)** Calculate the amount of I_2 remaining in the reaction mixture.

[2]

- (iv) Calculate the concentration of NH_2OH in 100 cm^3 sample of the photography developing solution. Hence conclude whether the solution is within the safety limit.

[2]

- (v) Iodate(V) ion, IO_3^- was used to generate iodine required for the reaction with NH_2OH .

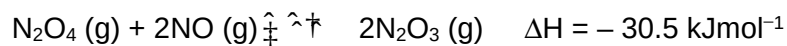
In this IO_3^- ion, the central iodine atom can expand its octet structure. Draw the dot-and-cross diagram of IO_3^- ion.

[1]

- (vi) From your answer in (a)(v), state the bond angle around the central iodine atom and account for the shape of IO_3^- ion.

[2]

- (b) In an industrial process, 5 mol of dinitrogen tetroxide, N_2O_4 and 7 mol of nitrogen monoxide, NO was heated in a 2.0 dm^3 vessel. The temperature is kept at 500°C . The two gases react slowly to form blue dinitrogen trioxide, N_2O_3 according to the following equation. It was found that the equilibrium mixture contains 2.3 mol of NO.



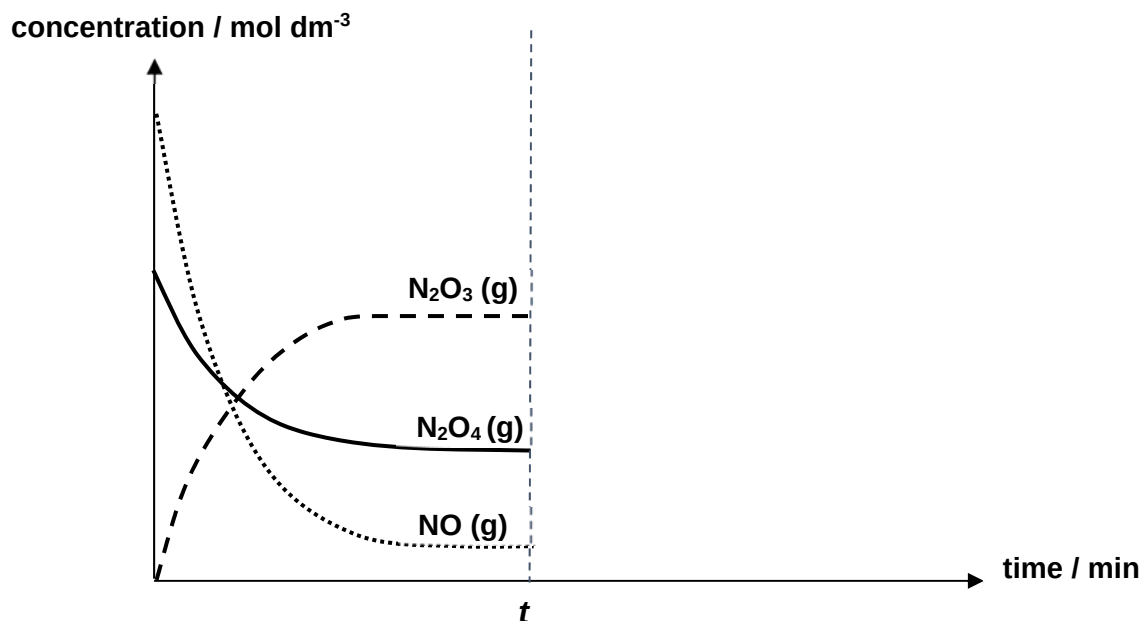
- (i) Write an expression for the equilibrium constant, K_c for the reaction, stating the units. [1]

- (ii) Determine the equilibrium concentrations of the three gases. Hence, calculate the value of K_c . [2]

- (iii) Predict and explain the effect of increasing temperature on the equilibrium position and equilibrium composition. [2]

- (iv) Sketch on the same axes, how the concentration of the three gases will change when temperature was increased at t .

[2]



- (v) State the operating pressure and temperature (high or low) required to obtain a high yield of N_2O_3 . Explain your answer.

[2]

- (vi) Hence, suggest why the operating temperature is fixed at a moderately high temperature of 500 °C.

[1]

[Total: 20]

END OF PAPER